

PROJECT REPORT

Import Data Using Transform Maps (Spreadsheet)

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Table of Contents

1. Introduction
2. Ideation Phase
3. Requirement Analysis
4. Project Design
5. Project Planning & Scheduling
6. Functional and Performance Testing
7. Results
8. Advantages & Disadvantages
9. Conclusion
10. Future Scope
11. Appendix

1. INTRODUCTION

1.1 Project Overview

Organisations often keep large amounts of data in spreadsheets such as Excel or CSV files. Entering this data into a platform record by record is slow and error-prone. This project, **Import Data Using Transform Maps (Spreadsheet)**, uses the ServiceNow import-set framework to load data from a spreadsheet into a staging (import set) table, and then uses a **Transform Map** to map each source column to the correct field of a target table. Options such as **coalesce** (to update existing records instead of creating duplicates) and transform scripts (to clean or convert values) ensure that the imported data is accurate and consistent.

1.2 Purpose

- To automate the loading of bulk data from spreadsheets into a ServiceNow table.
- To reduce manual data entry effort, time and human errors.
- To prevent duplicate records by using coalesce on a unique field such as Email.
- To give administrators a clear log of inserted, updated, ignored and failed records after every import.

2. IDEATION PHASE

2.1 Problem Statement

Administrators and data-entry staff receive employee, customer or asset details as spreadsheets and must type them into the system one by one. This takes hours, causes typing mistakes, creates duplicate records and makes it hard to track what was updated. There is a need for a simple, repeatable way to import spreadsheet data directly into the target table with correct field mapping and error reporting.

Problem Statement	I am	I am trying to	But	Because	Which makes me feel
PS-1	A system administrator	Load hundreds of records from a spreadsheet into a table	Manual entry is slow and I create duplicates and typos	Data is typed record by record with no mapping or validation	Frustrated and worried about data quality
PS-2	A project team member	Update existing records with new spreadsheet values	I cannot tell which records already exist	There is no key-based matching (coalesce)	Unsure and under time pressure

2.2 Empathy Map Canvas

SAYS "I have 500 rows to enter by hand." "Please don't create duplicates." "Can we just upload the Excel file?"	THINKS Manual entry will take too long. One wrong value can break reports. There must be an automatic way.
DOES Copies and pastes cell values. Checks records manually after entry. Repeats the same work for each new file.	FEELS Stressed about deadlines. Anxious about mistakes. Relieved when data loads correctly.
PAINS Time-consuming data entry; typing errors; duplicate records; no clear log of what changed.	GAINS Fast bulk upload; correct field mapping; no duplicates; clear import summary; reusable transform map.

Figure 1: Empathy map of the administrator / data-entry user

2.3 Brainstorming

The team discussed several ways to bring spreadsheet data into the platform and then voted on the best idea.

Team Member	Idea Proposed	Feasibility
E Dilli Kumar	Use Import Sets with a Transform Map to load Excel data into the target table.	High
R Vinish	Add Coalesce on a unique field (Email) so existing records are updated, not duplicated.	High
K Jerina	Use Mapping Assist for quick, drag-and-drop field mapping.	High
B Mohanavalli	Write a transform script to clean values (trim text, convert date formats).	Medium

Team Member	Idea Proposed	Feasibility
P Sri Ragavan	Check Transform History after each run and create a sample template sheet for users.	High

Selected idea: Import spreadsheet data through an Import Set and a Transform Map with coalesce, supported by a transform script and a review of the Transform History.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Awareness	Preparation	Upload	Mapping	Transform	Review
User Action	Receives data in a spreadsheet	Cleans sheet and creates target table	Uploads .xlsx/.csv via Load Data	Creates Transform Map and maps fields	Runs the transform	Opens Transform History and target table
Thinking	"Do I have to type this?"	"Are my columns correct?"	"Will the file load?"	"Are fields matched right?"	"Will duplicates appear?"	"Did all rows import?"
Feeling	Worried	Neutral	Hopeful	Focused	Anxious	Satisfied
Pain Point	Large manual effort	Inconsistent headers	Wrong file type	Unmapped fields	Duplicate or failed rows	Unclear error messages
Opportunity	Offer bulk import	Provide a template sheet	Validate file type	Mapping Assist	Coalesce + scripts	Clear counts and logs

Table: Customer journey of the administrator importing data

3.2 Solution Requirements

Functional Requirements

FR No.	Requirement	Description
FR-1	Target table creation	Create the target table with the required fields.
FR-2	Spreadsheet upload	Upload .xlsx / .csv file into an import set (staging) table.
FR-3	Transform Map	Create a Transform Map between the staging and target tables.
FR-4	Field mapping	Map every source column to a target field (auto or manual).
FR-5	Coalesce	Match records on a unique field to insert or update.
FR-6	Run transform	Execute the transform and load the data into the target table.
FR-7	Result review	Show inserted, updated, ignored and error counts in Transform History.

Non-Functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	Simple steps that a non-technical admin can follow.
NFR-2	Security	Only users with admin / import roles can run imports.
NFR-3	Reliability	Invalid rows must not stop valid rows from importing.
NFR-4	Performance	Import of 1000+ rows should complete without timeout.
NFR-5	Availability	Runs on the cloud-hosted ServiceNow instance.
NFR-6	Scalability	Same Transform Map can be reused for new files.

3.3 Data Flow Diagram

The diagram shows how data moves from the user's spreadsheet to the target table and how results are logged.

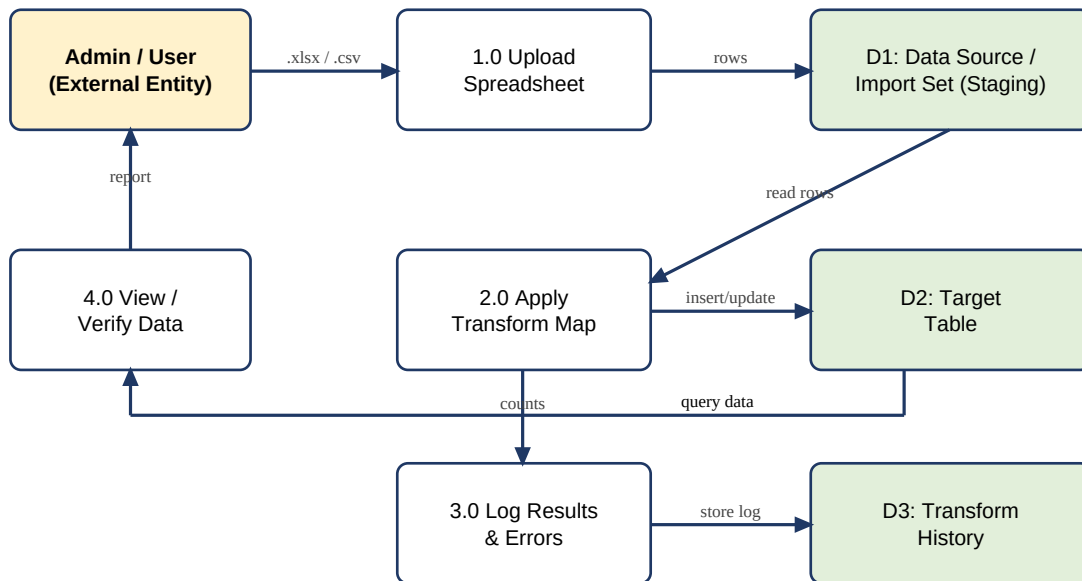


Figure 2: Data Flow Diagram (Level 1)

3.4 Technology Stack

Component	Technology / Tool
Platform	ServiceNow (Personal Developer Instance)
Data source	Microsoft Excel (.xlsx) / CSV files
Staging	Import Set table (Load Data)
Transformation	Transform Map, Field Maps, Coalesce, Transform Script (JavaScript)
Target storage	Custom table created in ServiceNow
Monitoring	Transform History and Import Set Rows
Testing	Manual UAT with sample spreadsheets

4. PROJECT DESIGN

4.1 Problem Solution Fit

Area	Description
Customer segment	Administrators, HR / data-entry staff and project teams who handle bulk data in spreadsheets.
Problem	Manual data entry is slow, error-prone and creates duplicates.
Existing alternatives	Typing records one by one or copy-pasting into list views.
Trigger	A new spreadsheet of records arrives and must be loaded quickly.
Our solution	Upload the spreadsheet once; the Transform Map automatically places each value into the right field.
Outcome	Faster, accurate and repeatable data loading with a clear audit log.

4.2 Proposed Solution

S.No.	Parameter	Description
1	Problem to be solved	Slow, error-prone manual entry of spreadsheet data.
2	Idea / solution	Import via Import Set + Transform Map with coalesce and scripts.
3	Novelty	Reusable map; supports insert and update in one run; built-in error log.
4	Social impact	Saves staff time so they can focus on higher-value work.
5	Business model	Included in the platform; no extra licence for the import feature.
6	Scalability	Handles larger files and can be reused for any new table or spreadsheet.

4.3 Solution Architecture

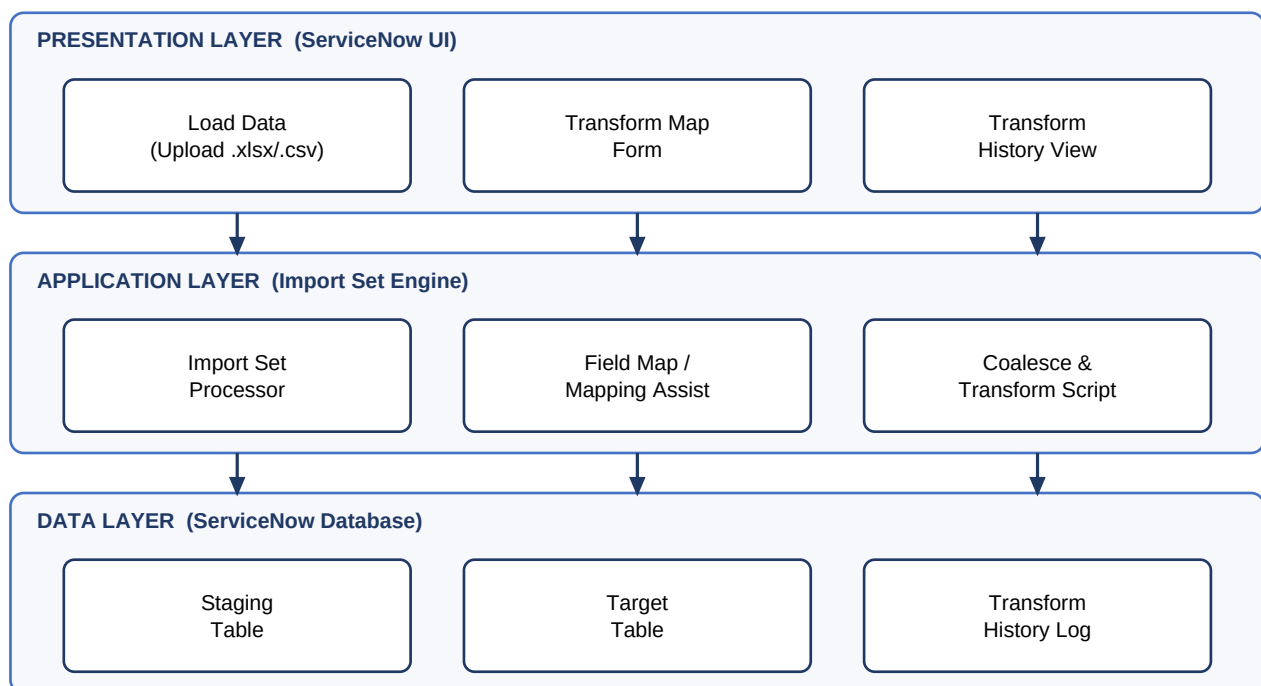


Figure 3: Solution architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	User Story / Task	Owner	Story Points	Duration
Sprint-1	Study import sets and transform maps; create the target table and fields	E Dilli Kumar	5	07 Sep – 12 Sep 2026
Sprint-1	Prepare sample spreadsheet data	R Vinish	3	07 Sep – 12 Sep 2026
Sprint-2	Upload spreadsheet and create the staging table	K Jerina	5	14 Sep – 19 Sep 2026
Sprint-2	Create Transform Map and field mappings	B Mohanavalli	8	14 Sep – 19 Sep 2026
Sprint-3	Configure coalesce and transform script	P Sri Ragavan	8	21 Sep – 26 Sep 2026
Sprint-3	Run transform and verify target data	E Dilli Kumar	5	21 Sep – 26 Sep 2026
Sprint-4	User Acceptance Testing and bug fixing	Entire team	8	28 Sep – 03 Oct 2026
Sprint-4	Documentation and final report	Entire team	5	28 Sep – 03 Oct 2026

Total story points: 47. Team velocity: about 12 story points per sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Functional testing was done through UAT: 14 test cases (positive and negative) were executed and all 14 passed, with 3 of 4 reported bugs closed. Performance was checked by importing spreadsheets of different sizes and noting the time taken.

S.No.	Number of Rows	Rows Inserted	Rows Failed	Approx. Time	Result
1	10	10	0	< 5 sec	Pass
2	100	100	0	< 10 sec	Pass
3	500	500	0	< 30 sec	Pass
4	1000	1000	0	< 60 sec	Pass

Note: Times are approximate and depend on the instance load.

7. RESULTS

7.1 Output Screenshots

Insert the screenshots of your own project in the boxes below.

[Screenshot placeholder]

Figure 4: Target table and fields

[Screenshot placeholder]

Figure 5: Spreadsheet loaded into the staging (import set) table

[Screenshot placeholder]

Figure 6: Transform Map with field mappings and coalesce

[Screenshot placeholder]

Figure 7: Transform History showing inserted/updated records

[Screenshot placeholder]

Figure 8: Final data in the target table

8. ADVANTAGES & DISADVANTAGES

Advantages

- Saves time by loading hundreds or thousands of records in one run.
- Reduces typing mistakes and improves data quality.
- Coalesce prevents duplicate records and allows updates.
- The Transform Map is reusable for future files.
- Transform History gives a clear audit trail.
- No external tools or coding are needed for basic imports.

Disadvantages

- Spreadsheet column names and formats must be consistent for correct mapping.
- Complex data conversion needs transform scripts (JavaScript knowledge).
- Error messages for blank or wrongly formatted data can be unclear.
- Very large files may need to be split or scheduled.
- Depends on the ServiceNow platform and user roles.

9. CONCLUSION

The project successfully demonstrated how data from an Excel/CSV spreadsheet can be imported into a ServiceNow table using an Import Set and a Transform Map. Field mapping, coalesce and transform scripts ensured accurate, duplicate-free data, and the Transform History provided clear results. In UAT, all 14 test cases passed and only one low-severity issue remained open. The solution greatly reduces manual effort and is easy to reuse.

10. FUTURE SCOPE

- Schedule automatic imports from a shared location (Scheduled Import / MID Server).
- Import data directly from web services, databases or REST APIs.
- Add validation scripts with clear, user-friendly error messages.
- Provide a self-service portal page to upload spreadsheets with a downloadable template.
- Send email notifications with the import summary after every run.
- Build dashboards to analyse import success and failure trends.

11. APPENDIX

Item	Details
Source Code	Transform script (if any) – attach the script used in the Transform Map / field maps
Dataset Link	[Add link to the sample spreadsheet, e.g., Google Drive link]
GitHub Link	[Add your GitHub repository link]
Project Demo Link	[Add your demo video link]

Sample dataset columns: Name, Email, Department, Phone, Date of Joining

Sample transform script (onBefore):

```
(function runTransformScript(source, map, log, target) {  
  // remove extra spaces and keep email in lower case  
  target.name = source.u_name.toString().trim();  
  target.email = source.u_email.toString().trim().toLowerCase();  
})(source, map, log, target);
```